

Student:	Email:
Jalen Salgado	

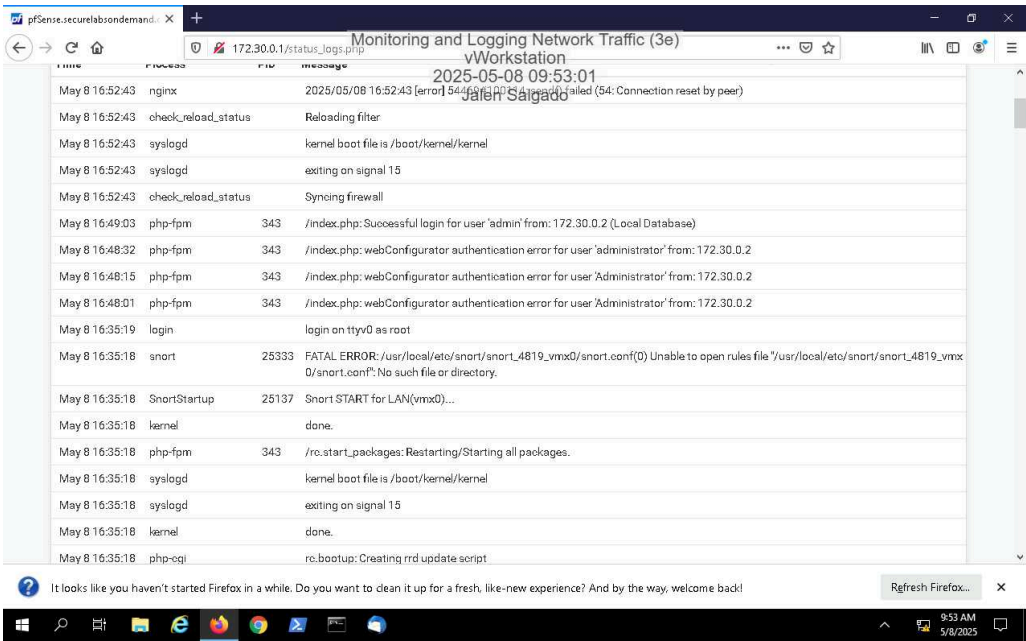
Time on Task:	Progress:
2 hours, 20 minutes	100%

Report Generated: Saturday, July 26, 2025 at 2:11 PM

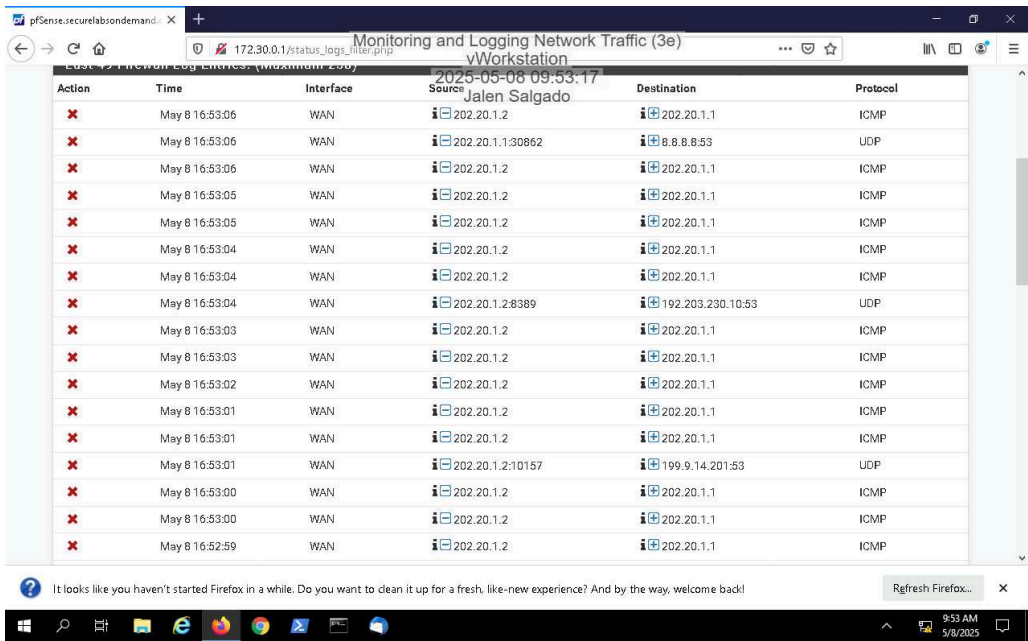
Section 1: Hands-On Demonstration

Part 1: Configure the pfSense Firewall Log

- 13. Make a screen capture showing the system logs.

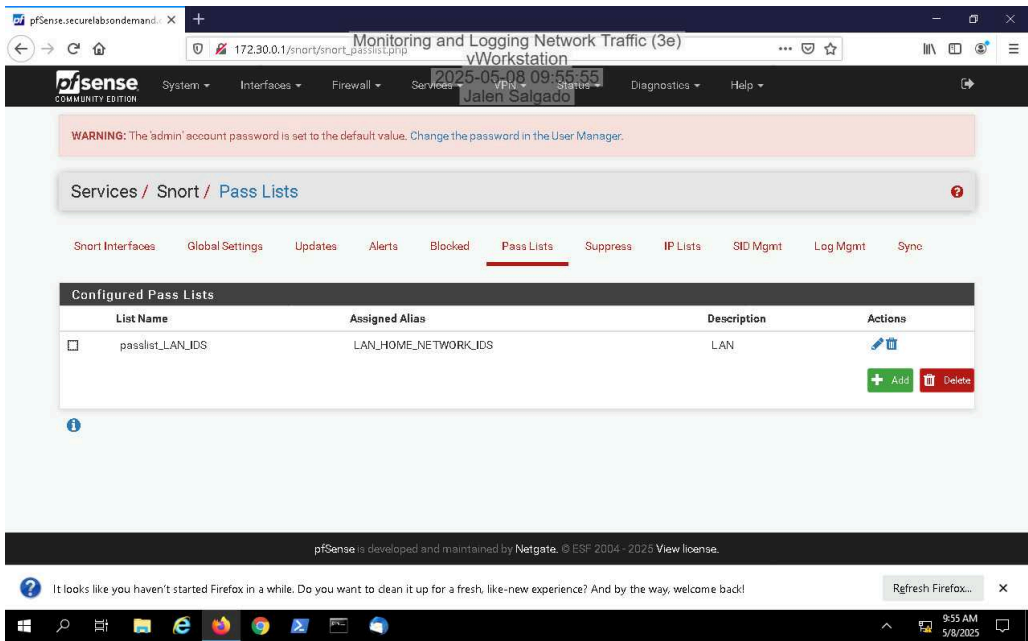


15. Make a screen capture showing the firewall logs.

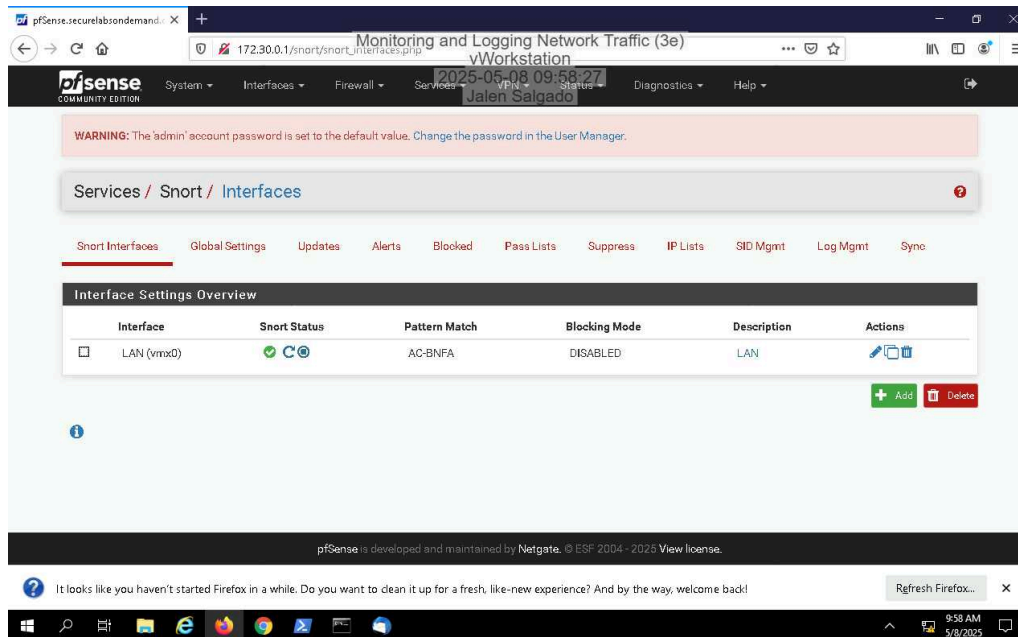


Part 2: Configure a Snort Intrusion Detection System

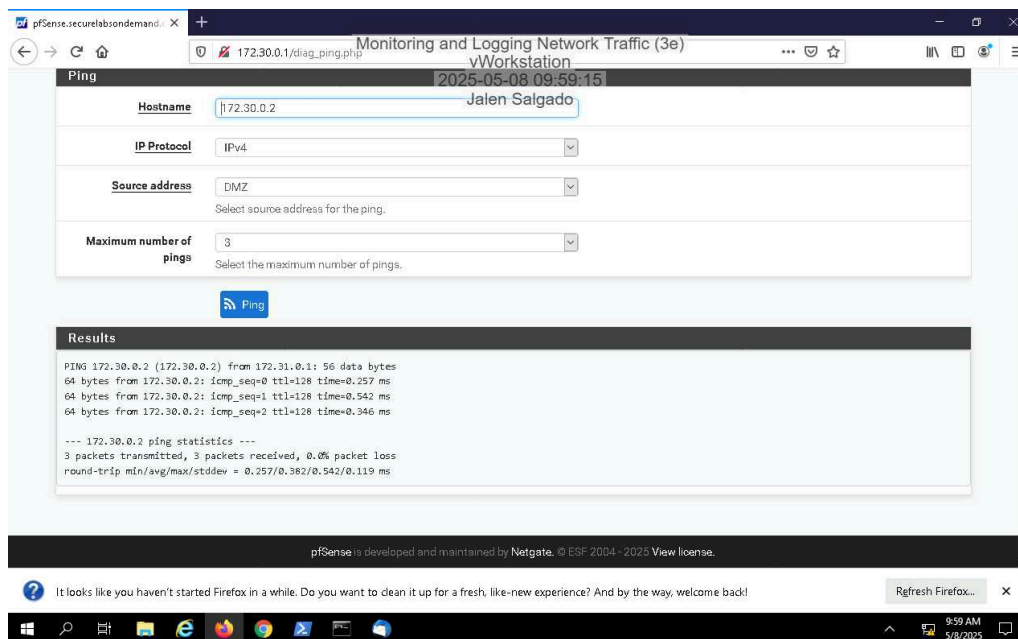
14. Make a screen capture showing the updated Pass Lists page.



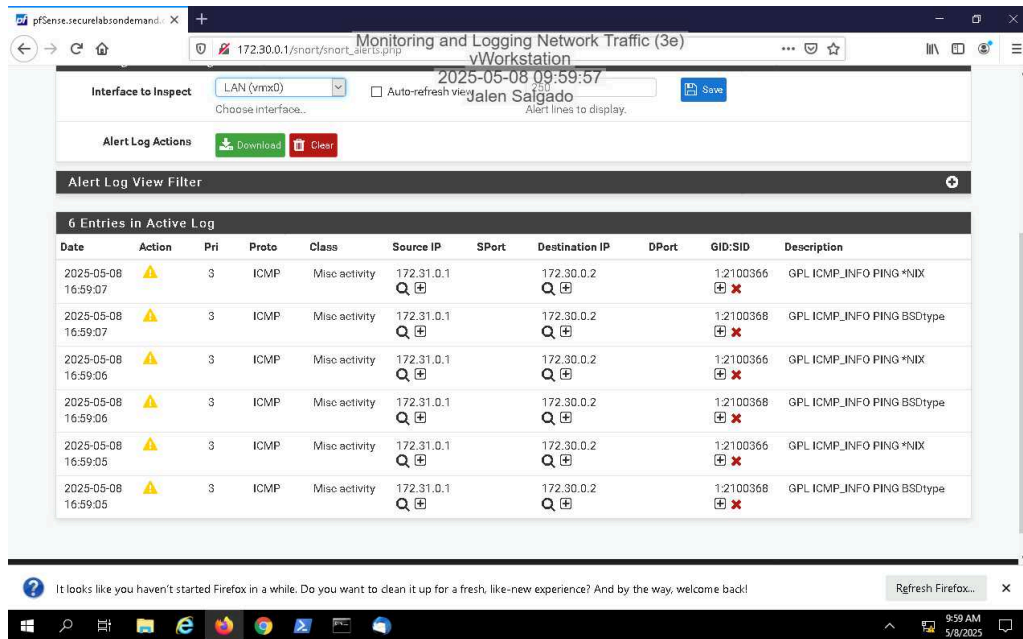
28. Make a screen capture showing the active Snort status on the LAN interface.



33. Make a screen capture showing the successful ping results.

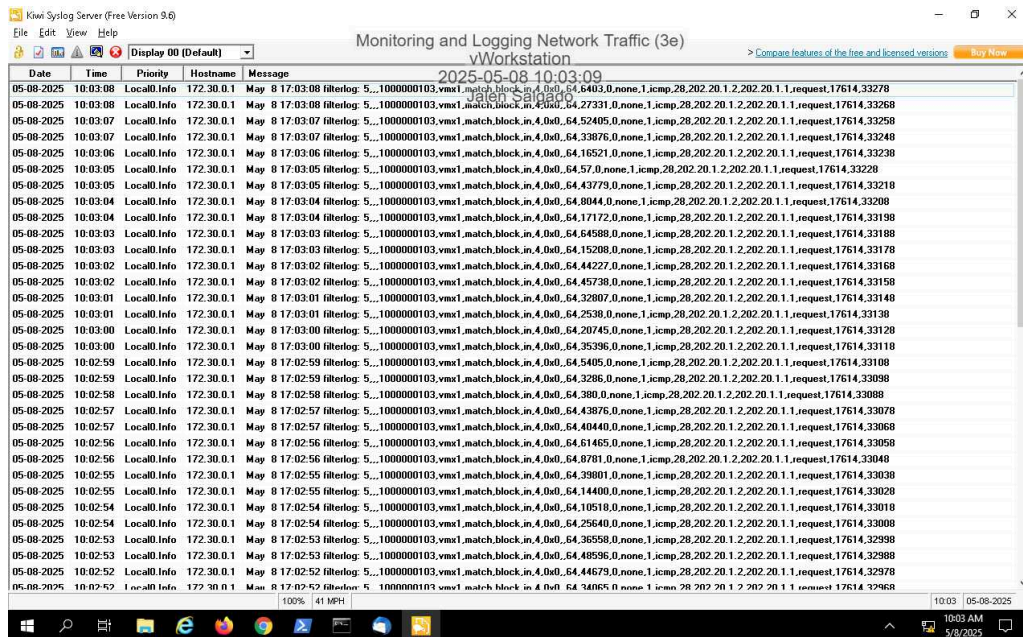


38. Make a screen capture showing the ICMP alerts in the Snort Active Log.



Part 3: Implement Firewall Log Forwarding with Kiwi Syslog Server

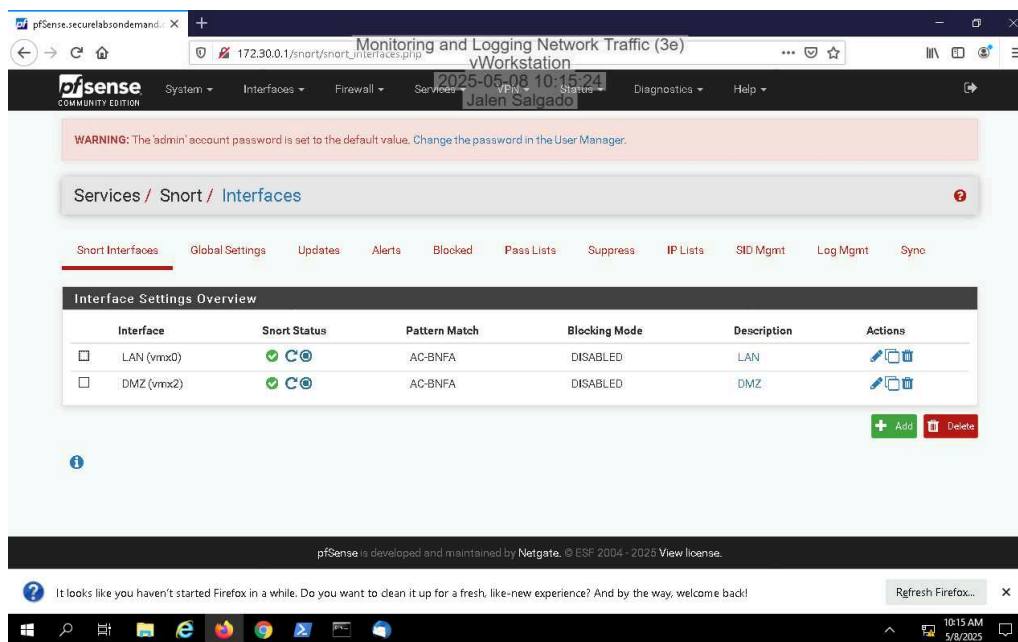
17. Make a screen capture showing the pfSense firewall log events in Kiwi Syslog Server.



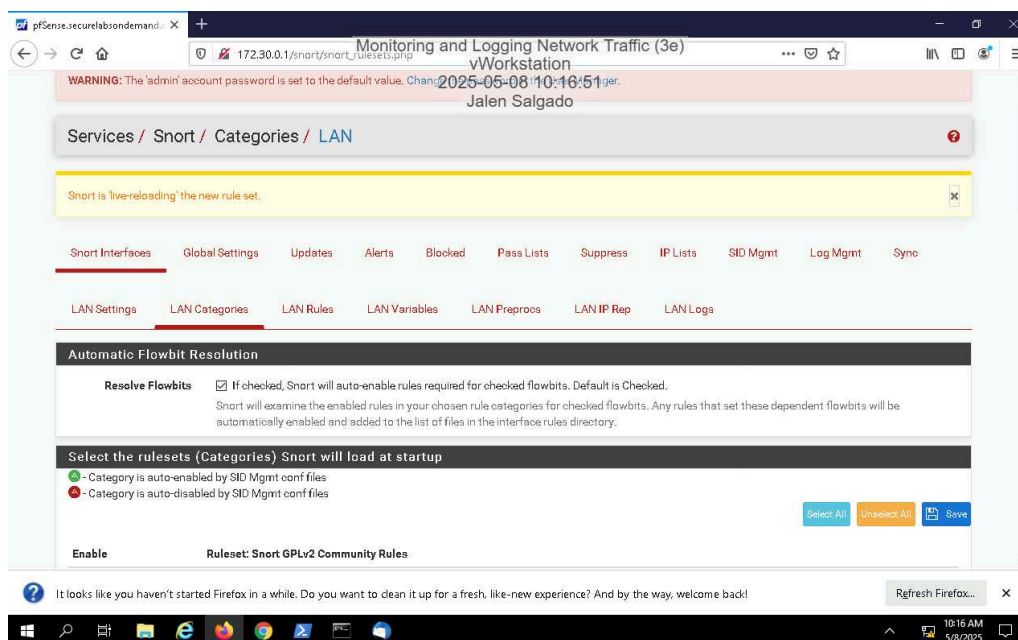
Section 2: Applied Learning

Part 1: Configure Snort Monitoring on the DMZ

17. Make a screen capture showing the active Snort status on the DMZ interface.

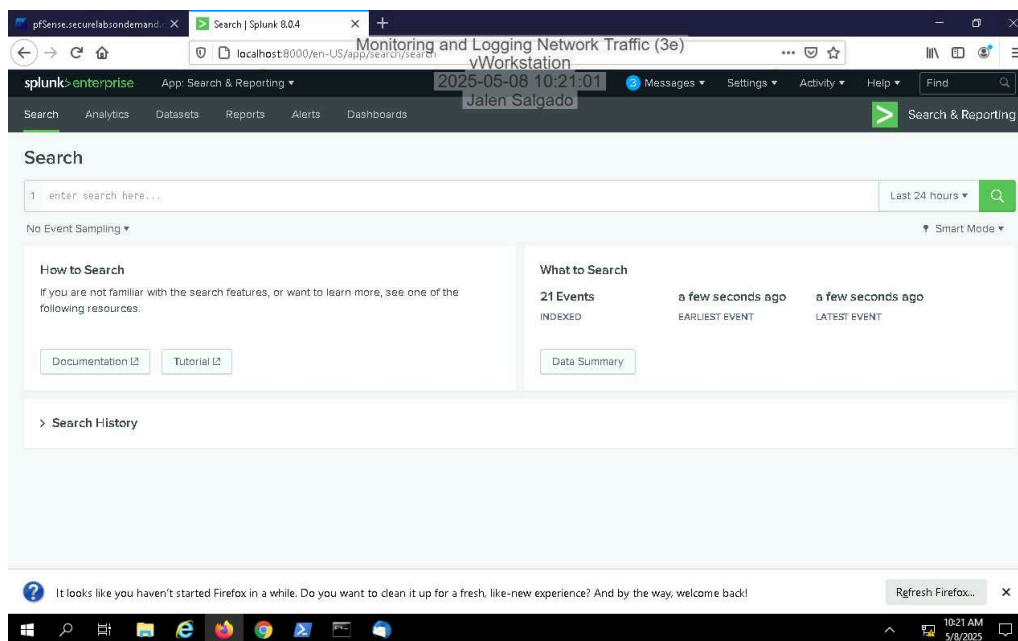


20. Make a screen capture showing the Snort GPLv2 Community Rules enabled and "live-reloading" message.



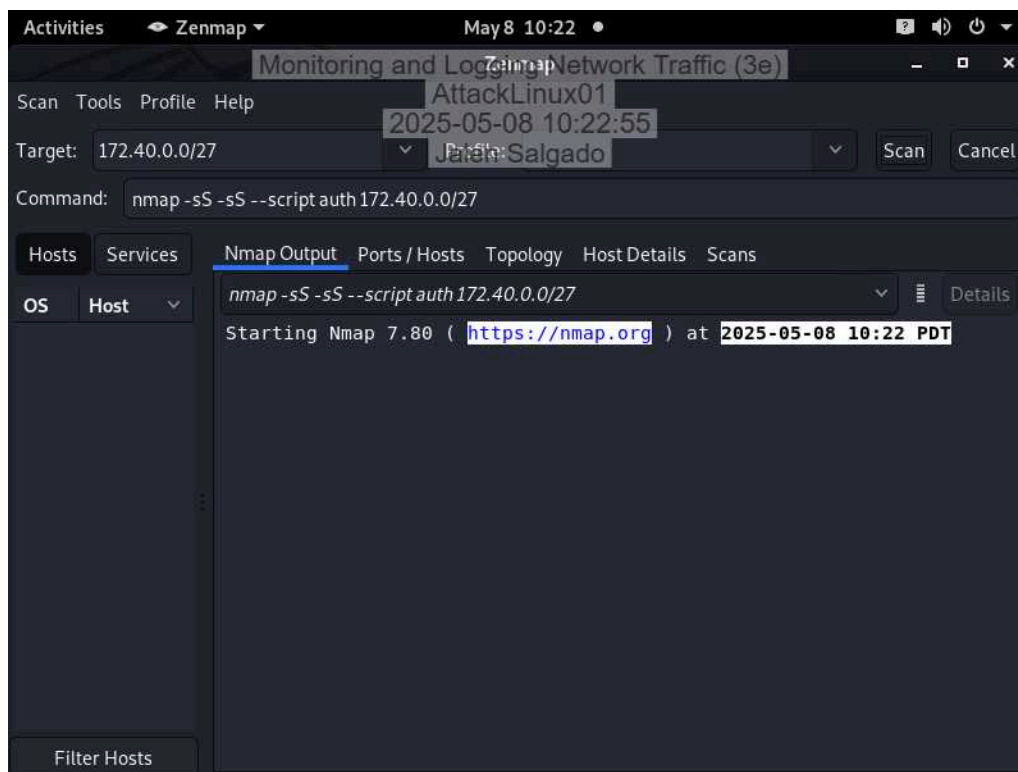
Part 2: Implement Security Information and Event Management with Splunk

13. Make a screen capture showing the indexed events in Splunk.

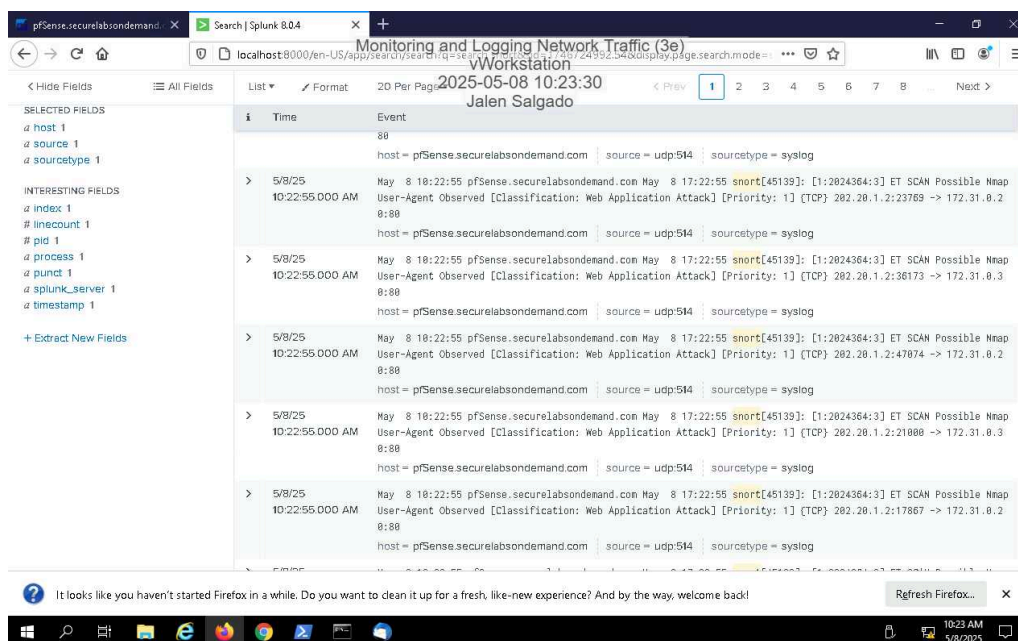


Part 3: Simulate and Detect a Perimeter Network Attack

6. Make a screen capture showing the Nmap scan report.



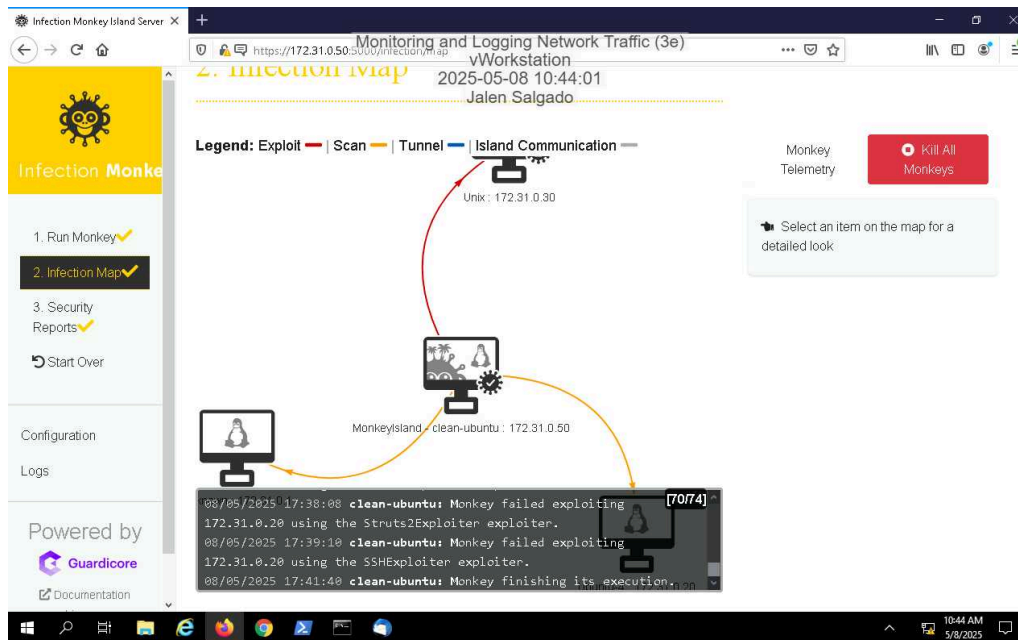
9. Make a screen capture showing the search results in Splunk.



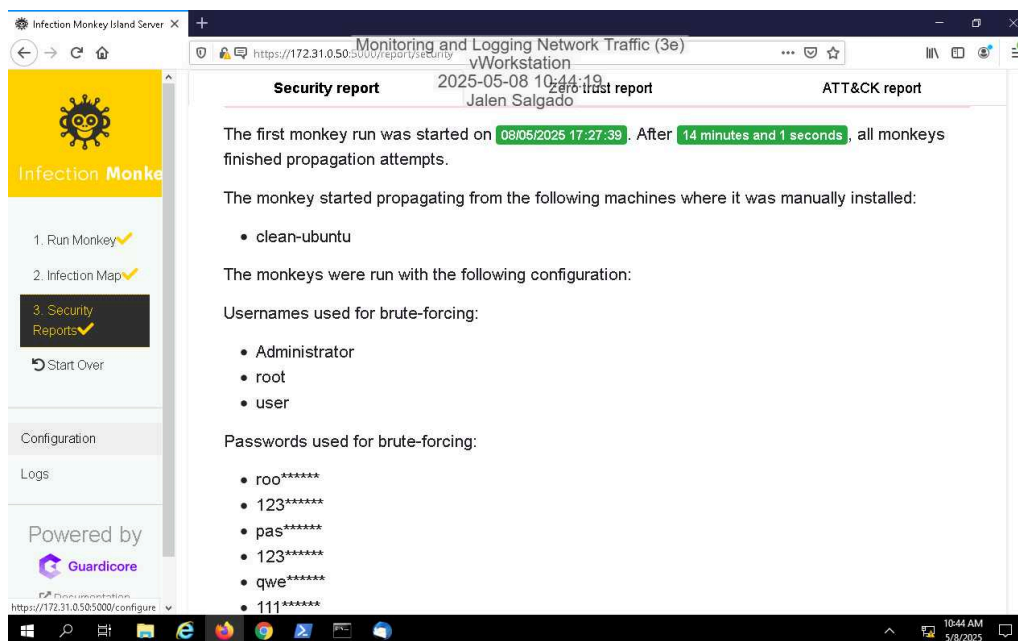
Section 3: Challenge and Analysis

Part 1: Simulate a DMZ Breach with Infection Monkey

Make a screen capture showing the resulting Infection Map.



Make a screen capture showing the resulting Security Report.

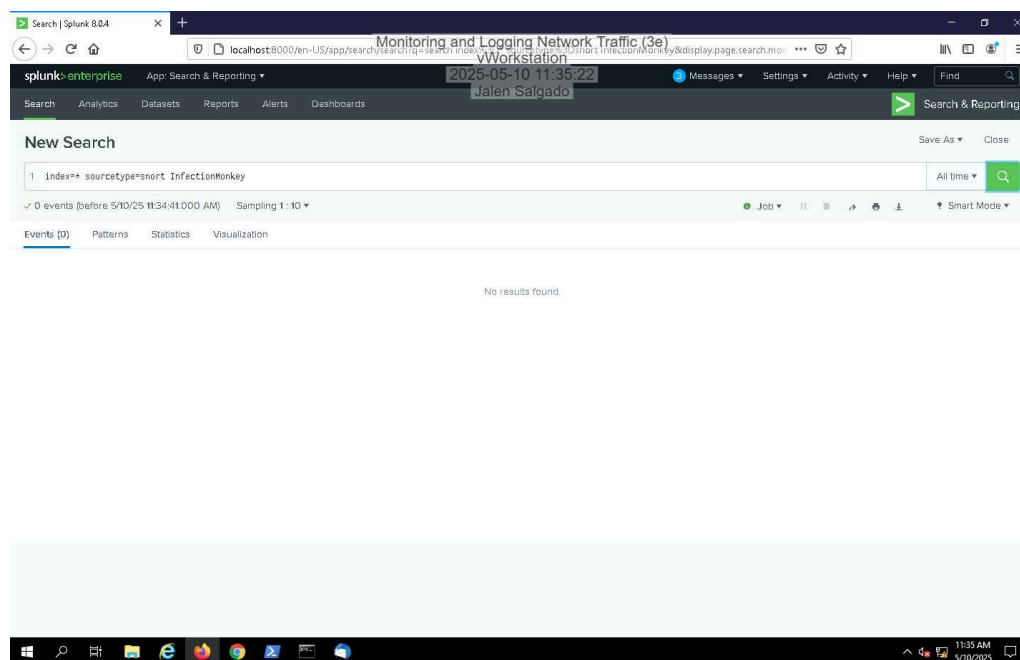


Summarize your DMZ breach simulation results, highlighting what you found to be the greatest concerns from a network monitoring perspective.

alot of the vulnerabilites came from using the basic username and password settings also immediate threats were also from things that werent update properly

Part 2: Detect a Simulated DMZ Breach with Snort and Splunk

Make a screen capture showing the results of your search query for Infection Monkey traffic in Splunk.



Describe any concerns about the structure of the query result or the data elements it contains. What data fields would you add, remove, or edit to make log analysis more effective?

The main concern is that no results were returned despite running a known attack simulation. This could be due to Snort not logging traffic correctly or Splunk not ingesting the logs. Even with logs present, many Snort alerts lack structured fields like destination IP, protocol, or severity level, which limits effective analysis. I would recommend including fields such as threat category, asset tags, and normalized timestamps to make the logs easier to parse and correlate during investigations

Write a brief memo to your manager describing Splunk's usefulness in detecting traces of your simulated breach. What configuration changes would you recommend? How would you enhance its functionality?

During the Infection Monkey simulation, I attempted to retrieve related Snort alerts in Splunk. Despite correctly querying for events, no results were returned. This indicates either a lack of logging from Snort or an ingestion/configuration issue with Splunk. To improve detection, I recommend reviewing Snort's output settings and ensuring Splunk is monitoring the correct log source. Additionally, enriching logs with structured fields like severity, IPs, and attack type would enhance detection accuracy. With the right configurations, Splunk can be a powerful tool for identifying and responding to simulated or real threats.